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Assignment - 3

Question # 1

Temp = 1 if it greater or equal 30°C
Press = 1 if it $\geq 50\text{psi}$

M.A = Alarm active if it is 1 (warning

Task A:

Truth Table

when Alarm
is on or both T
and P is on)

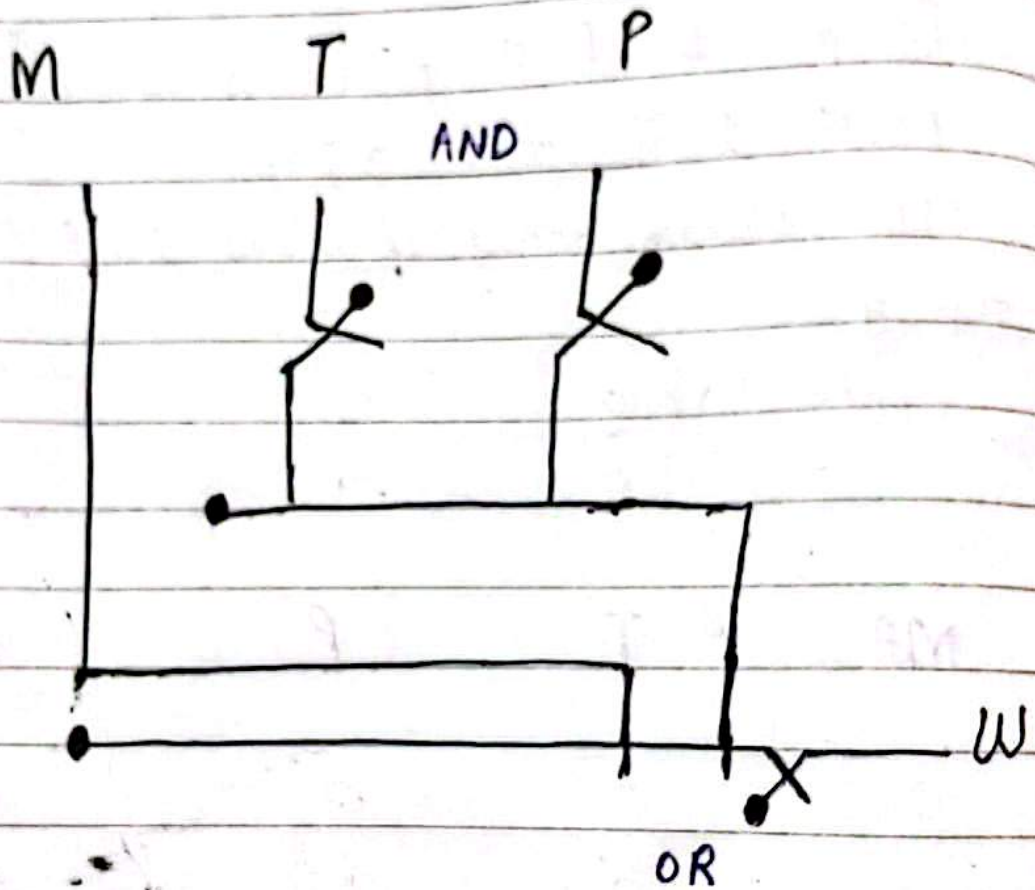
MA	T	P	Warning
0	0	0	0
0	0	1	0
0	1	0	0
0	1	1	1
1	0	0	1
1	0	1	1
1	1	0	1
1	1	1	1

Task B : Boolean Expression

$$\text{Warning} = M + T \cdot P$$

①

logic circuit diagram



Question # 2

Automatic Sump pump

L = water level = 1 if it's greater than 10 cm

R = Heavy rain = 1 if expected rain.

B = Backup = 1 if battery fully charged

output pump is 1 if water level = 1
it's also pump when rain and backup
both on

Truth Table

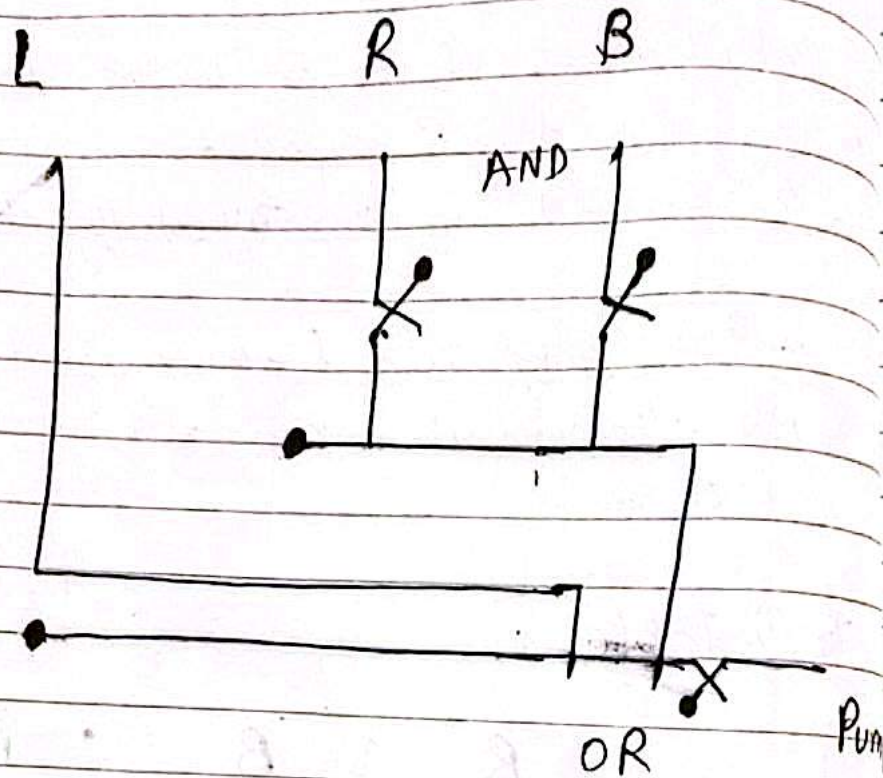
L	R	B	P
0	0	0	0
0	0	1	0
0	1	0	0
0	1	1	1
1	0	0	1
1	0	1	1
1	1	0	1
1	1	1	1

Boolean Expression

$$P = L + R \cdot B$$

②

logic circuit diagram



Question # 3

S = server temp = 1 if $> 80^{\circ}\text{C}$

M = Maintenance = 1 if switch active

A = ambient room = 1 if room temperature is $> 25^{\circ}\text{C}$

Output cooling Fan it's on if $S=1$ and $M=0$

it also = 1 if $S=1$ and $A=1$

Q3

Truth Table

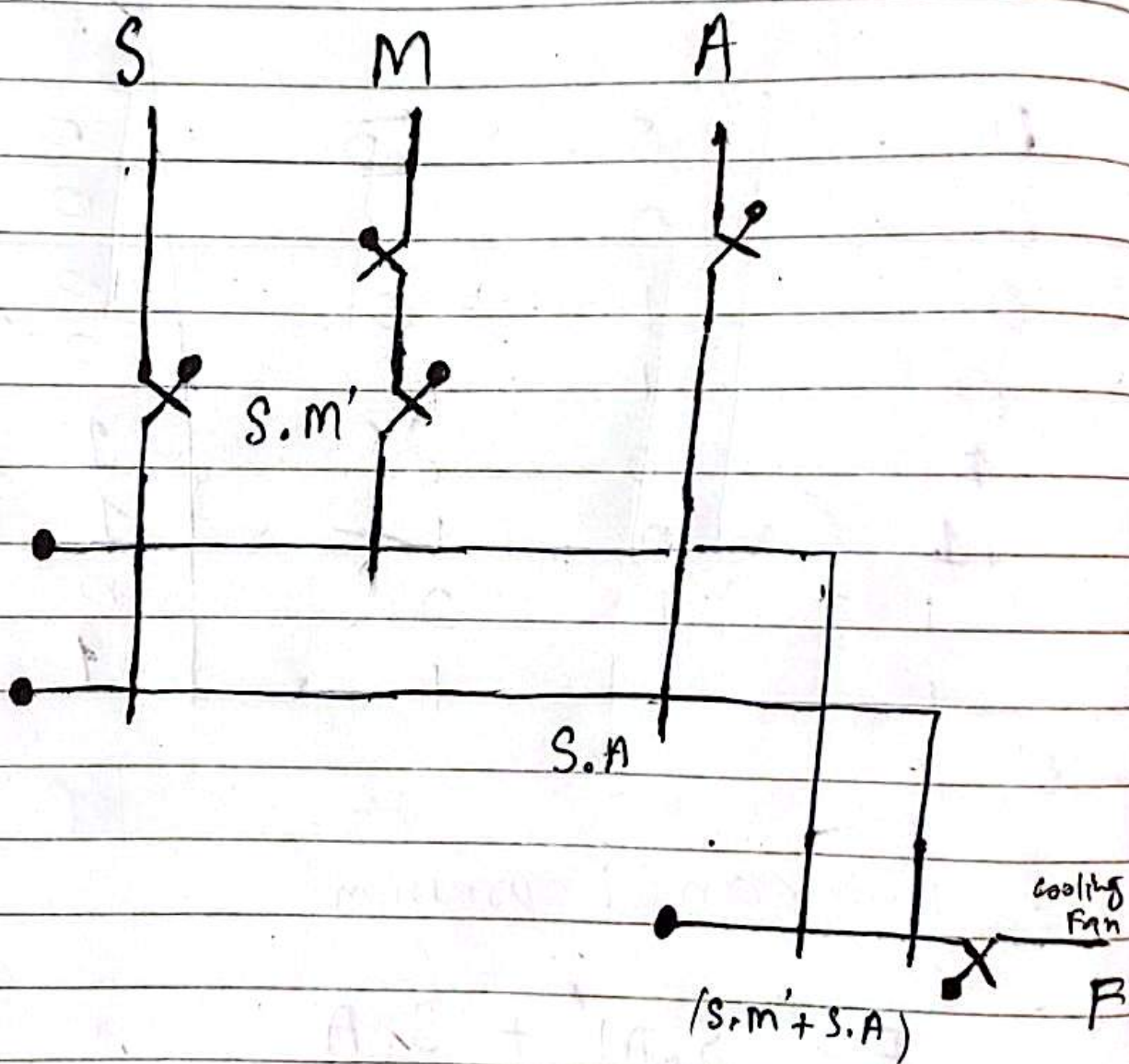
S	M	A	F
0	0	0	0
0	0	1	0
0	1	0	0
0	1	1	0
1	0	0	1
1	0	1	1
1	1	0	0
1	1	1	1

Boolean Expression

$$F = S.M' + S.A$$

③

logic circuit diagram



Q # 4

$T = \text{Time lock} = 1$ if time is b/w 9am-5pm
 $Mg = \text{Manager's key} = 1$ when inserted
 $Cs = \text{Cashier's key} = 1$ when inserted

output vault lock = 1 if ~~and~~ T and at least one key is inserted

Truth Table

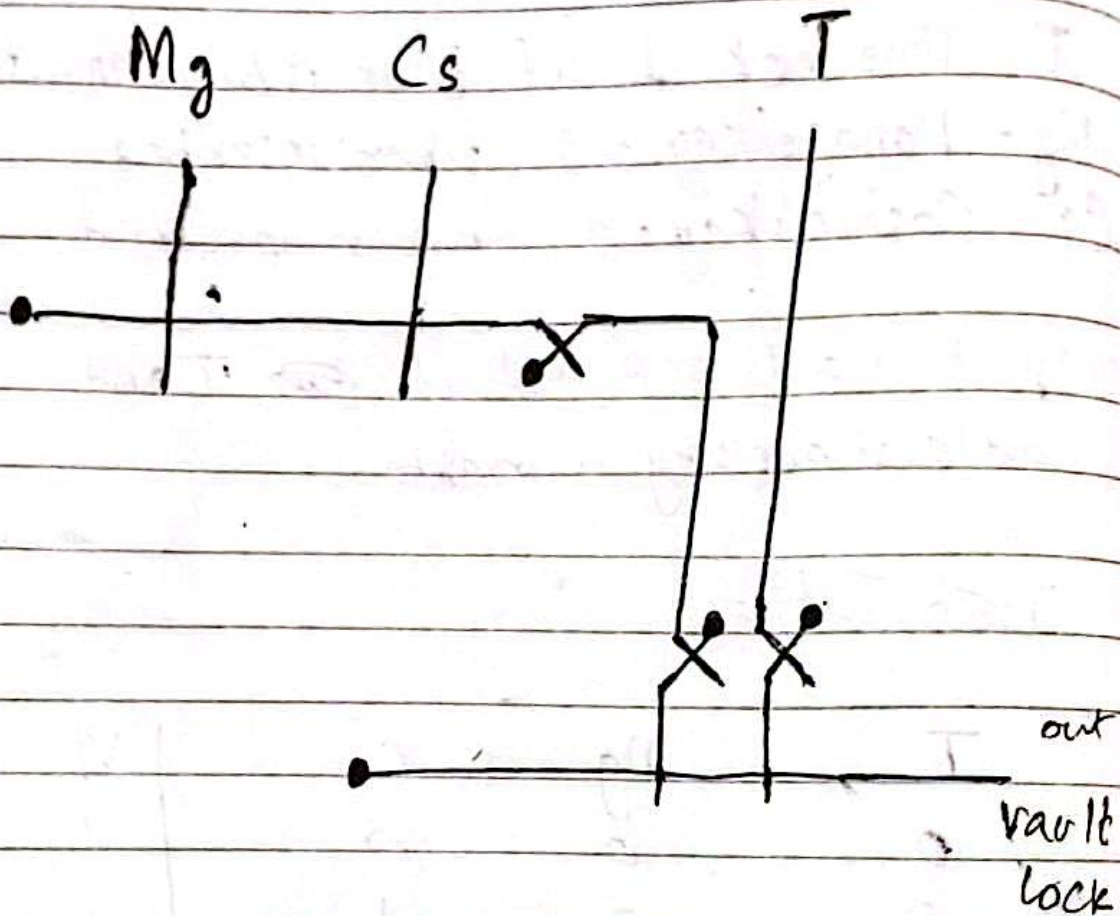
T	Mg	Cs	VL
0	0	0	0
0	0	1	0
0	1	0	0
0	1	1	0
1	0	0	0
1	0	1	1
1	1	0	1
1	1	1	1

Boolean expression:

$$VL = (Mg + Cs) \cdot T$$

④

logical circuit



Question # 5

$T = 1$ if outside temperature $< 4^{\circ}\text{C}$

$H = 1$ if cabin humidity $> 70\%$

$B = 1$ if defrost button pressed

Truth Table

T	H	B	D
0	0	0	0
0	0	1	0
0	1	0	0
0	1	1	0
1	0	0	0
1	0	1	0
1	1	0	0
1	1	1	0

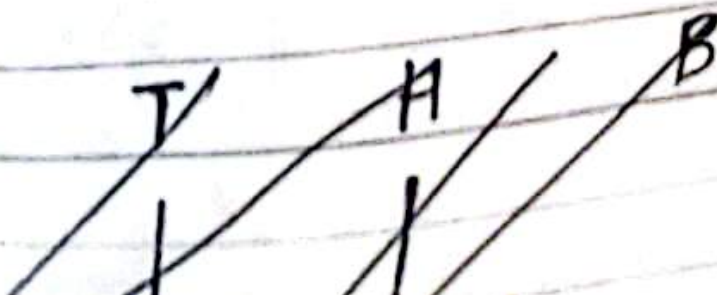
Q5

Truth Table

T	H	B	D
0	0	0	0
0	0	1	1
0	1	0	0
0	1	1	1
1	0	0	0
1	0	1	1
1	1	0	1
1	1	1	1

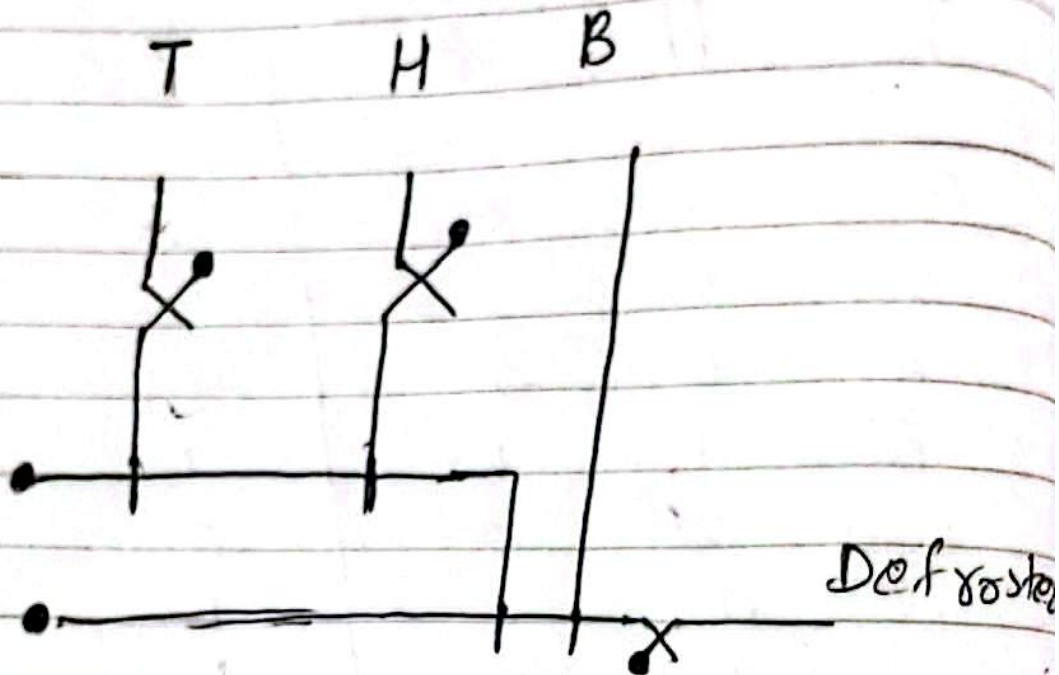
Boolean Expression

$$D = T \cdot H + B$$



(5)

logical circuit diagram



Question # 6

$W_t = 1$ if package weight $> 5\text{ kg}$
 $H_t = 1$ if package too tall $> 50\text{ cm}$
 $M_t = 1$ if metal detected

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Truth Table

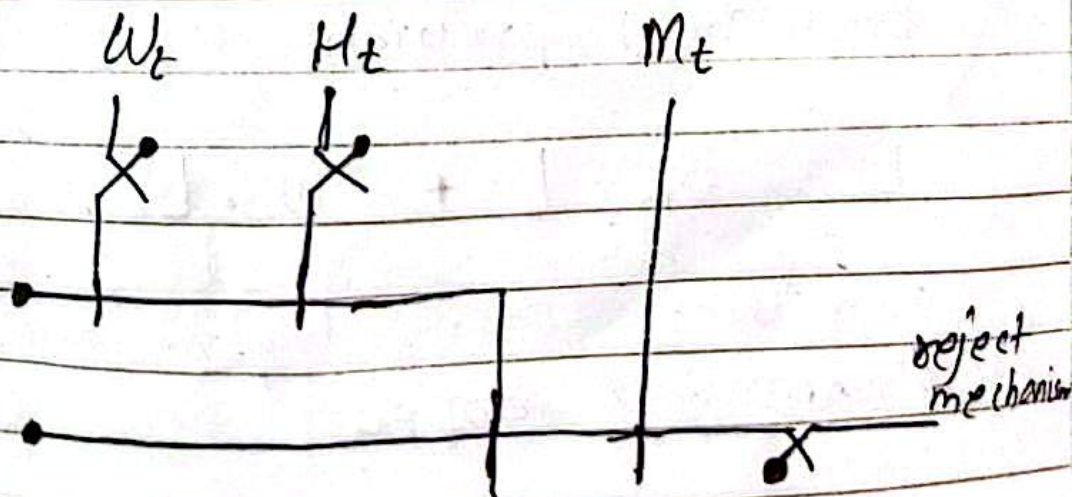
W_t	H_t	M_t	R
0	0	0	0
0	0	1	1
0	1	0	0
0	1	1	1
1	0	0	0
1	0	1	1
1	1	0	1
1	1	1	1

Boolean Expression:

$$R = M_t + W_t \cdot H_t$$

metal defect alone is rejected

⑥ logical circuit diagram



Question # 7

$B_m = 1$ if Mechanical Bumper struck

$W_m = 1$ if weight limit exceeded

$L = 1$ if laser wire broken

Truth Table

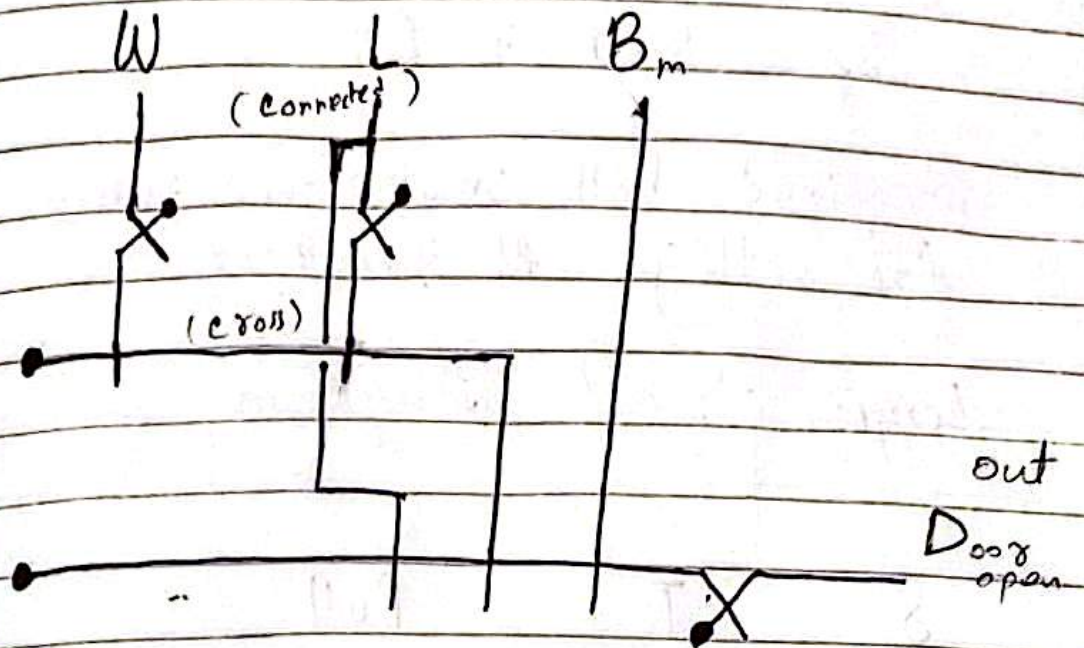
B_m	W_m	L	D
0	0	0	0
0	0	1	1
0	1	0	0
0	1	1	1
1	0	0	1
1	0	1	1
1	1	0	1
1	1	1	1

Boolean Expression

$$D = B_m + L + W_m \cdot L$$

expression clearly explain when elevator door reopen

⑦



Question # 8

$S = 1$ if smoke $> 20\%$
 $T = 1$ if ceiling temp $> 75^\circ\text{C}$
 $P_{\text{pull}} = 1$ if manual pull active

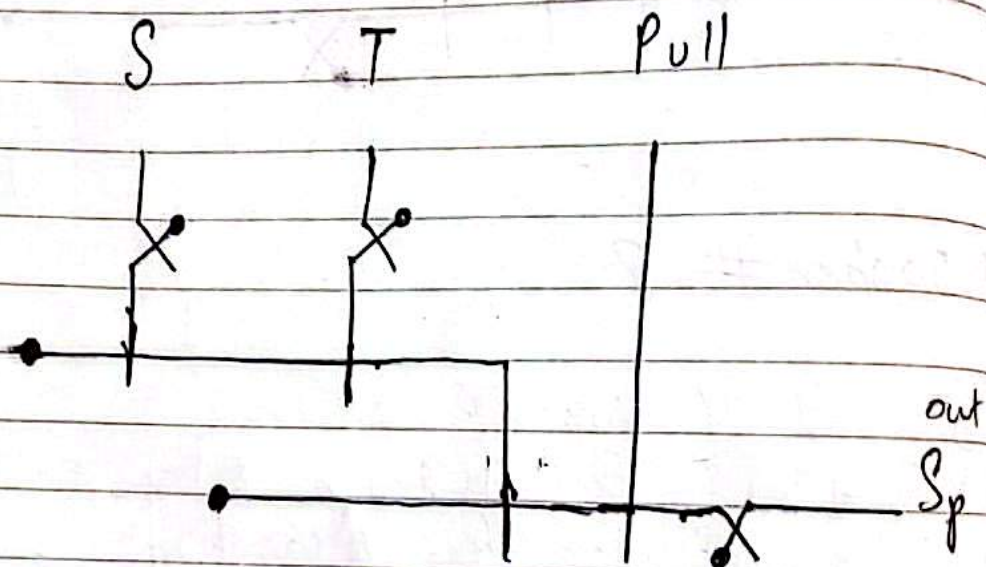
S	T	P _{pull}	S _p
0	0	0	0
0	0	1	1
0	1	0	0
0	1	1	1
1	0	0	0
1	0	1	1
1	1	0	1
1	1	1	1

Q8

Boolean Expression
 $S_p = S \cdot T + Pull$

Required both smoke and temp
for setting off sprinkler.

Logic circuit diagram



Q9

Auto chicken coop door

$L = 1$ if Daylight > 500 lux
 $M = 1$ if A chicken inside
 $R = 1$ if currently raining

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Truth Table

L	M	R	D
0	0	0	0
0	0	1	0
0	1	0	1
0	1	1	1
1	0	0	0
1	0	1	0
1	1	0	0
1	1	1	1

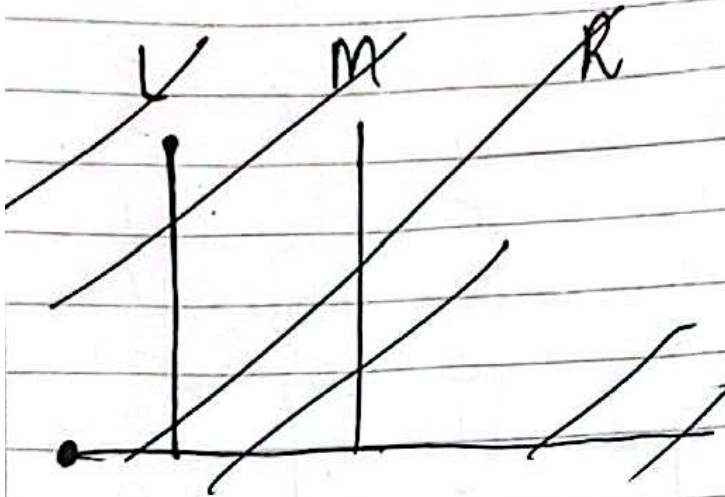
Boolean Expression

$$D = L' \cdot M + L \cdot R \cdot M$$

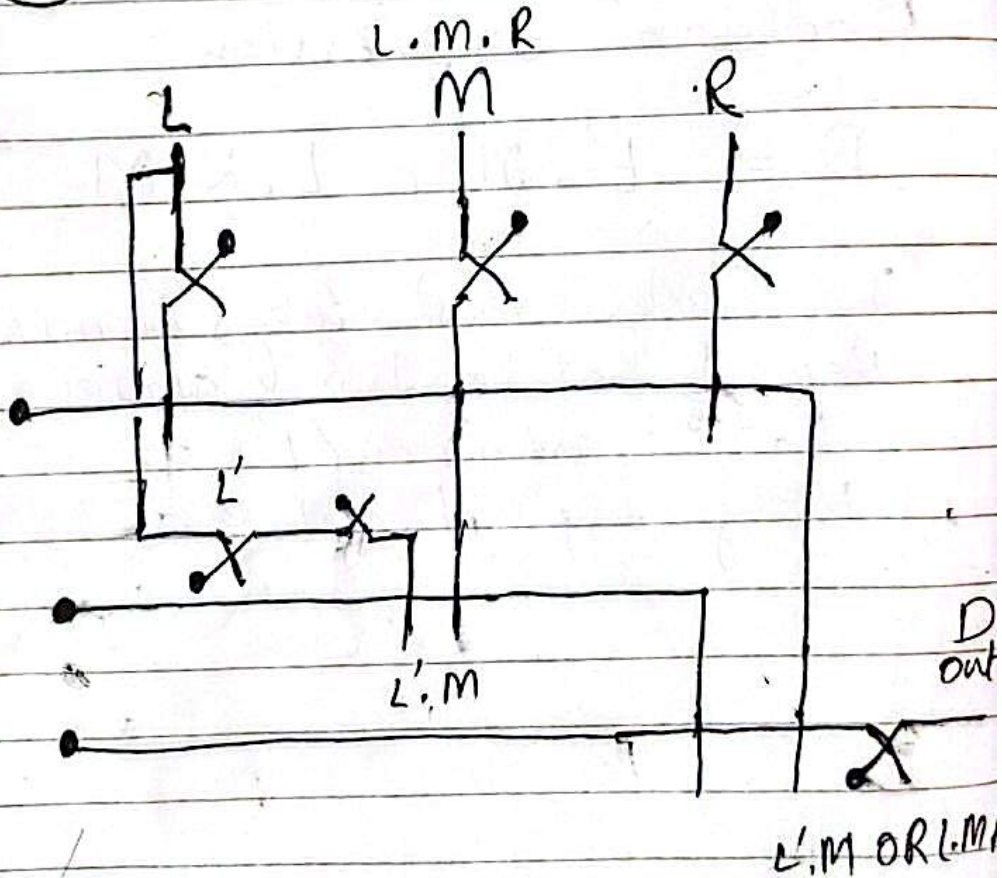
The chicken must always be inside.
 Beyond that the door closes
 at night means (L') or
 during day is ($L \cdot R$).

Q4

(a) logic circuit diagram



(b) circuit



Question # 10

$B = 1$ if Battery $< 15\%$.

$D = 1$ if Distance from
Launch pad $> 2\text{ km}$

$W = 1$ if Wind speed $> 20\text{ km/h}$

Truth Table

B	D	W	S
0	0	0	0
0	0	1	0
0	1	0	0
0	1	1	0
1	0	0	0
1	0	1	1
1	1	0	1
1	1	1	1

Boolean expression

$$S = B \cdot D + W \cdot B$$

Q 10

logical circuit Diagram

